CYLINDER HEAD

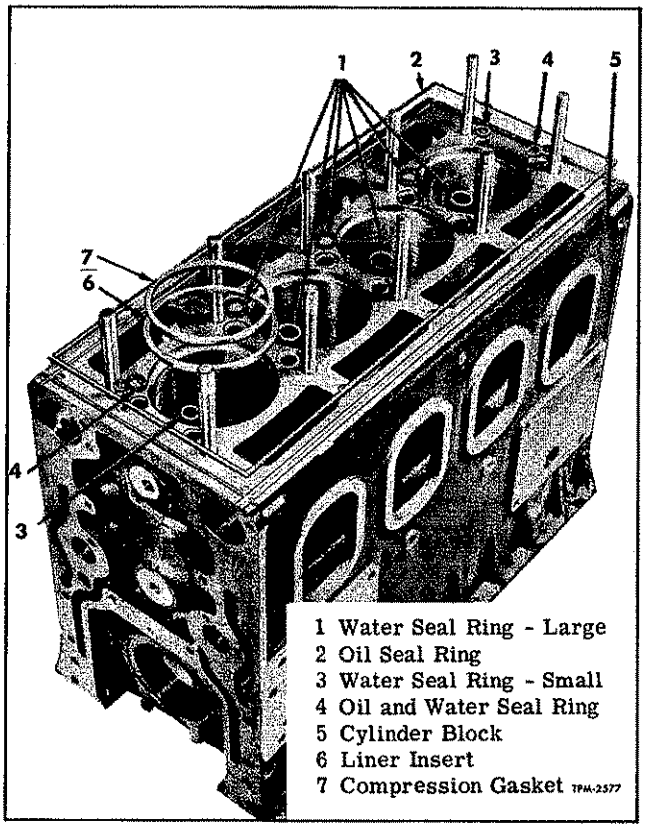


Figure 72—Cylinder Compression Gaskets, Oil and Water Seals

3. Loosen jumper line nut at injector, then remove line at connectors and swing line aside so that cylinder head stud nuts can be reached.

4. Remove cylinder head hold-down stud nuts using special wrench to reach nuts under injector control tube. If wrench is not available remove control tube and brackets.

CAUTION: **When removing cylinder head with injectors in place, care must be taken to avoid damaging injector tips and cam followers at the instant head clears the cylinder head studs. Raise cylinder head clear of studs. DO NOT ALLOW CYLINDER HEAD TO BIND ON STUDS.**

5. When resting cylinder head assembly on work bench, protect injector tips, also cam followers and rollers by resting the valve side of head on wooden blocks at least 2″ thick.

6. Remove cylinder head compression and oil seal gaskets.

7. If engine is removed from vehicle for a major overhaul, and is positioned in engine overhaul stand, cylinder liner clamps should be installed between each two cylinders to prevent possibility of liners falling from block while engine is turned to various positons in overhaul stand. Install liner clamps over cylinder block studs and secure with cylinder head hold down stud nuts.

DISASSEMBLY

Conditions will arise when a complete cylinder head overhaul becomes necessary. For such conditions the removal of all working parts attached to cylinder head is necessary. NOTE: New cylinder heads are furnished with valve guides, studs, injector copper tubes and plugs. If a new cylinder head is to be installed, some of the used parts removed from the old head, or new parts must be installed in the head before it is attached to cylinder block. The following procedure applies when a cylinder head is to be completely overhauled:

1. Remove rocker arms and shafts as instructed under "Rocker Arm, Shaft and Bracket Replacement" in VALVES AND INJECTOR OPERATING MECHANISM section later in this manual.

2. Remove valve springs as instructed under "Valve Spring Replacement" in VALVES AND INJECTOR OPERATING MECHANISM section later in this manual. After valve springs and locks are removed valves may be removed from lower side of cylinder head.

3. Remove injectors as instructed under FUEL INJECTORS in **FUEL SYSTEM** section of this manual.

4. Remove fuel oil connectors and manifolds as instructed under "Fuel Manifolds" in **FUEL SYSTEM** section of this manual.

5. Remove cam follower and guide as instructed under "Push Rod, Cam Follower, and Spring Replacement" in VALVES AND INJECTOR OPERATING MECHANISM section later in this manual.

CLEANING

Whenever engine is overhauled, the cylinder head should be thoroughly inspected for any condition that would render it unfit for further service. Such inspection will take place after the head has been removed from the block, completely disassembled, and thoroughly cleaned in either live steam or a suitable solvent; then cleaned with dry compressed air.

If shop facilities permit, grease and scale should be removed from disassembled head in following manner: